

Qinghao Hu

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Experience

- 2024.08– **Massachusetts Institute of Technology**
Present - Postdoc in Department of Electrical Engineering and Computer Science
- Advisor: Prof. Song Han
- 2024.02– **ETH Zürich**
2024.04 - Academic Visitor in ETH System Group
- Advisor: Prof. Ana Klimovic
- 2024.01– **Nanyang Technological University**
2024.08 - Research Assistant Professor

Education

- 2020–2023 **Nanyang Technological University**
- Ph.D. in Computer Science
- Advisor: Prof. Tianwei Zhang and Prof. Yonggang Wen
- 2018–2020 **National University of Singapore**
- S.M. in Electrical and Computer Engineering
- Advisor: Prof. Kelvin Fong
- 2014–2018 **Zhejiang University**
- B.Eng. in Electronic Information Science and Technology
- Advisor: Prof. Wenyan Yin

Awards & Honors

- 2023 **Google Ph.D. Fellowship** (*Only-ever* recipient from Singapore in Systems & Networking)
- 2023 **Distinguished Paper Award** of ASPLOS '23
- 2023 **Best Paper Award** of WAIC '23
- 2024 **Rising Star** in ML and Systems
- 2024 **Best Ph.D. Thesis Award** of Nanyang Technological University (College of Computing)
- 2018 **Best Undergraduate Thesis Award** of Zhejiang University
- 2018 **Outstanding Graduate** of Zhejiang University

Publications

Summary: I have published 9 first/co-first authored papers in top-tier system and machine learning conferences/journals like OSDI, NSDI, ASPLOS, ATC, SC, ICLR. As of March 2026, my work has received over 1,322 citations on Google Scholar, with an h-index of 15 and an i10-index of 18.

* : Equal Contribution, † : System Lead, ‡ : My Mentee
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- ASPLOS '26 **Taming the Long-Tail: Efficient Reasoning RL Training with Adaptive Drafter**
Qinghao Hu*, Shang Yang^{*‡}, Junxian Guo, Xiaozhe Yao[‡], Yujun Lin, Yuxian Gu, Han Cai, Chuang Gan, Ana Klimovic, Song Han
Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2026
- OSDI '26 **LLMFabric: Heterogeneous and Decentralized Large-Scale Machine Learning Serving System**
Xiaozhe Yao[‡], Youhe Jiang, Ilia Badanin, **Qinghao Hu**, Binhang Yuan, Imanol Schlag, Eiko Yoneki, Ana Klimovic
USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2026
- CVPR '26 **ForeAct: Steering Your VLA with Efficient Visual Foresight Planning**
Zhuoyang Zhang, Shang Yang[‡], **Qinghao Hu**, Luke J. Huang[‡], James Hou, Yufei Sun, Yao Lu, Song Han
Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- NeurIPS '25 **Jet-Nemotron: Efficient Language Model with Post Neural Architecture Search**
Yuxian Gu, **Qinghao Hu**[†], Shang Yang[‡], Haocheng Xi, Junyu Chen, Song Han, Han Cai
Neural Information Processing Systems (NeurIPS), 2025
- NeurIPS '25 **Scaling up Reasoning to Long Videos in VLMs**
Yukang Chen, Wei Huang, Baifeng Shi, **Qinghao Hu**[†], Hanrong Ye, Ligeng Zhu, Zhijian Liu, Pavlo Molchanov, Jan Kautz, Xiaojuan Qi, Sifei Liu, Hongxu Yin, Yao Lu, Song Han
Neural Information Processing Systems (NeurIPS), 2025
- MLSys '25 **LServe: Efficient Long-sequence LLM Serving with Unified Sparse Attention**
Shang Yang^{*‡}, Junxian Guo*, Haotian Tang, **Qinghao Hu**, Guangxuan Xiao, Jiaming Tang, Yujun Lin, Zhijian Liu, Yao Lu, Song Han
Conference on Machine Learning and Systems (MLSys), 2025
- ICLR '25 **LongVILA: Scaling Long-Context Visual Language Models for Long Videos**
Yukang Chen*, Fuzhao Xue*, Dacheng Li*, **Qinghao Hu**^{*†}, Ligeng Zhu, Xiuyu Li, Yunhao Fang, Haotian Tang, Shang Yang[‡], Zhijian Liu, Ethan He, Hongxu Yin, Pavlo Molchanov, Jan Kautz, Linxi Fan, Yuke Zhu, Yao Lu, Song Han
International Conference on Learning Representations (ICLR), 2025
- EuroSys '25 **DeltaServe: Multi-Tenant Language Model Serving via Delta Compression**
Xiaozhe Yao[‡], **Qinghao Hu**, Ana Klimovic
European Conference on Computer Systems (EuroSys), 2025
- NSDI '24 **Characterization of Large Language Model Development in the Datacenter**
Qinghao Hu*, Zhisheng Ye*, Zerui Wang^{*‡}, Guoteng Wang, Meng Zhang, Qiaoling Chen, Peng Sun, Dahua Lin, Xiaolin Wang, Yingwei Luo, Yonggang Wen, Tianwei Zhang
USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2024
 **Invited Submission to USENIX ;login:**
- CSUR '24 **Deep Learning Workload Scheduling in GPU Datacenters: A Survey**
Zhisheng Ye*, Wei Gao*, **Qinghao Hu***, Peng Sun, Xiaolin Wang, Yingwei Luo, Tianwei Zhang, Yonggang Wen
ACM Computing Surveys (CSUR), 2024

- OSDI '23 **Hydro: Surrogate-Based Hyperparameter Tuning Service in Datacenters**
Qinghao Hu, Zhisheng Ye, Meng Zhang, Qiaoling Chen, Peng Sun, Yonggang Wen, Tianwei Zhang
USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2023
- ASPLOS '23 **Lucid: A Non-Intrusive, Scalable and Interpretable Scheduler for Deep Learning Training Jobs**
Qinghao Hu*, Meng Zhang*, Peng Sun, Yonggang Wen, Tianwei Zhang
Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2023
🏆 Distinguished Paper Award
- ATC '22 **Primo: Practical Learning-Augmented Systems with Interpretable Models**
Qinghao Hu, Harsha Nori, Peng Sun, Yonggang Wen, Tianwei Zhang
USENIX Annual Technical Conference (ATC), 2022
- SC '21 **Characterization and Prediction of Deep Learning Workloads in Large-Scale GPU Datacenters**
Qinghao Hu, Peng Sun, Shengen Yan, Yonggang Wen, Tianwei Zhang
Conference for High Performance Computing, Networking, Storage, and Analysis (SC), 2021
🏆 Top-3 Cited Paper in SC '21 (as of December 2025)

Open-Source Software

All of my research is open-source. Below are several projects that I have led or contributed to:

[LongVILA](#), 3700+ stars
[Jet-Nemotron](#), 700+ stars
[OminiServe](#), 700+ stars
[Long-RL](#), 600+ stars
[InternEvo](#), 400+ stars
[FastRL](#), 100+ stars
[Acme](#), 100+ stars
[Lucid](#), [Hydro](#), [Primo](#)

Professional Service

Workshop Organizer

- CVPR '25 Workshop: Efficient Large Vision Models Workshop (ELVM)
MICRO '24 Workshop: Hardware and Architectural Support for Security and Privacy (HASP)

Conference Reviewer

- ICLR '26 International Conference on Learning Representations
ICML '26 International Conference on Machine Learning
NeurIPS '26 Conference on Neural Information Processing Systems
ECCV '26 European Conference on Computer Vision
CVPR '26 Conference on Computer Vision and Pattern Recognition
ICLR '25 International Conference on Learning Representations
EuroSys '25 EuroSys Conference – Shadow Committee Member

EuroSys '24 EuroSys Conference – Shadow Committee Member
EuroSys '23 EuroSys Conference – Shadow Committee Member
OSDI '22 USENIX Operating Systems Design and Implementation – AE Committee Member
ATC '22 USENIX Annual Technical Conference – AE Committee Member
EuroSys '22 EuroSys Conference – AE Committee Member
SOSP '21 Symposium on Operating Systems Principles – AE Committee Member
Journal Reviewer
TPDS IEEE Transactions on Parallel and Distributed Systems
TACO ACM Transactions on Architecture and Code Optimization
TOCS ACM Transactions on Computer Systems
CSUR ACM Computing Surveys